



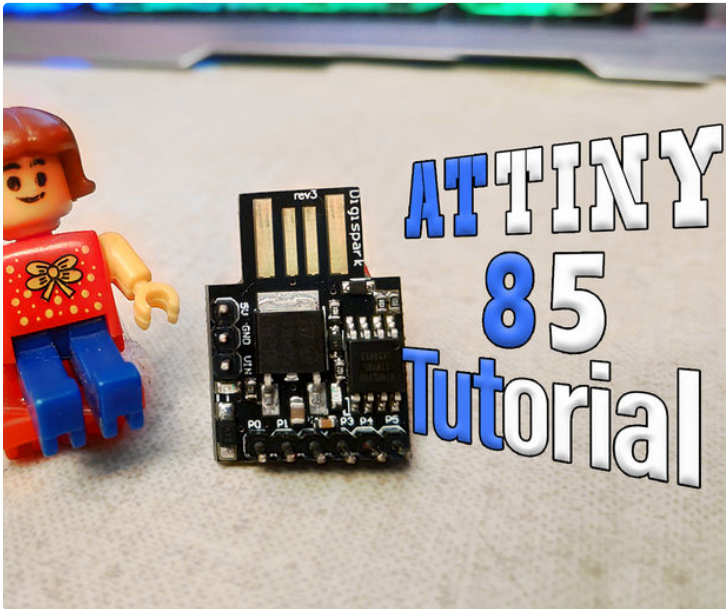
How to Setup DigiSprak Attiny85 Board



by MissionCritical

Hello there Engineers and Hobbyists, we all love Arduino's right! But using an Arduino UNO, NANO or any other Arduino for a small task like *displaying* a simple message on **LCD** or just *Fading LED* would be

waste of resource and money, so in this video, we would discover a **Cheap** and **Compact** Alternative of Arduino called **ATTINY85**, so let's get started!



Step 1: Watch the Video

[//www.youtube.com/embed/U3wj9dQt50Q](https://www.youtube.com/embed/U3wj9dQt50Q)

Step 2: Gathering the Requirements

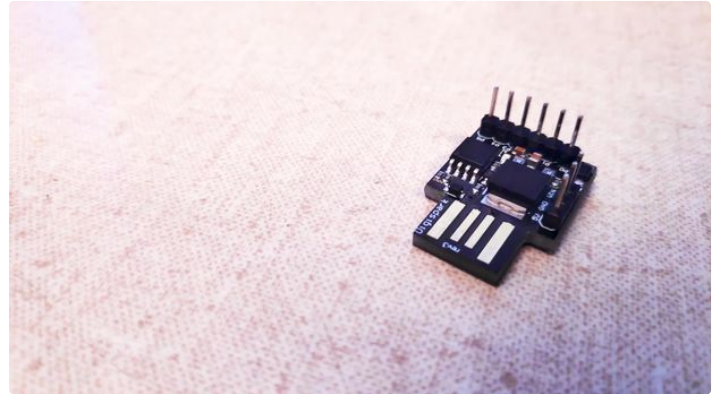
for this Tutorial, we will need

1. Attiny85 Board

https://in.gearbest.com/boards-shields/pp_00949109...

2. Arduino IDE

<https://www.arduino.cc/en/main/software>



```
BlinkCode | Arduino 1.8.8 (Windows Store 1.8.19.0)
File Edit Sketch Tools Help

BlinkCode

1 /* Blink Code by Akshay Momaya
2  * for Mission Critical Channel
3  * ATTINY85 Tutorial for <Mission
4  * subscribe for more arduino T
5  https://www.youtube.com/channel/
6  */
7
8 void setup()
9 {
10   pinMode(1, OUTPUT);
11 }
12
13
14 void loop()
15 {
16   digitalWrite(1, HIGH);
17   delay(1000);
18   digitalWrite(1, LOW);
19   delay(1000);
20 }
```

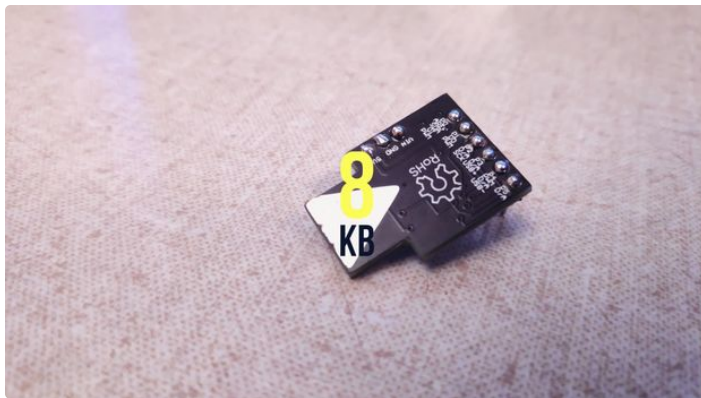
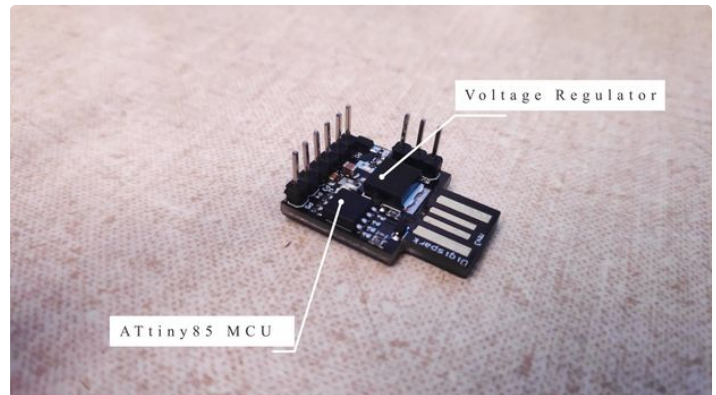
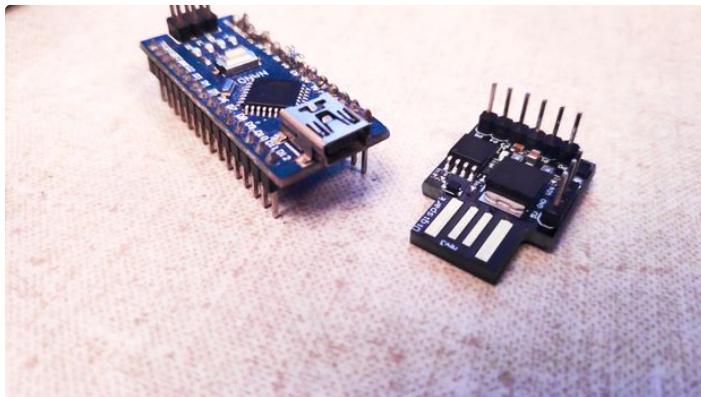
Step 3: Understanding the Board

At Tiny 85 is similar to Arduino, but it is not Arduino, ATTINY 85 is a Micro-Controller chip, which has 6 General Purpose IO pins(GPIO), out of which **5** are **PWM** enabled, and ATTINY85 also supports the **SPI** and **I2C** Communication Protocols, but it has only **8Kb** of programming memory, so yes, it is not as capable as Arduino board, but it will work just fine for small projects.

also, this board few other features like

- inbuilt 8 bit timers.
- inbuilt comparators.
- 512 Bytes EPROM
- 512 Bytes SRAM.

here is a complete data sheet for brief knowledge



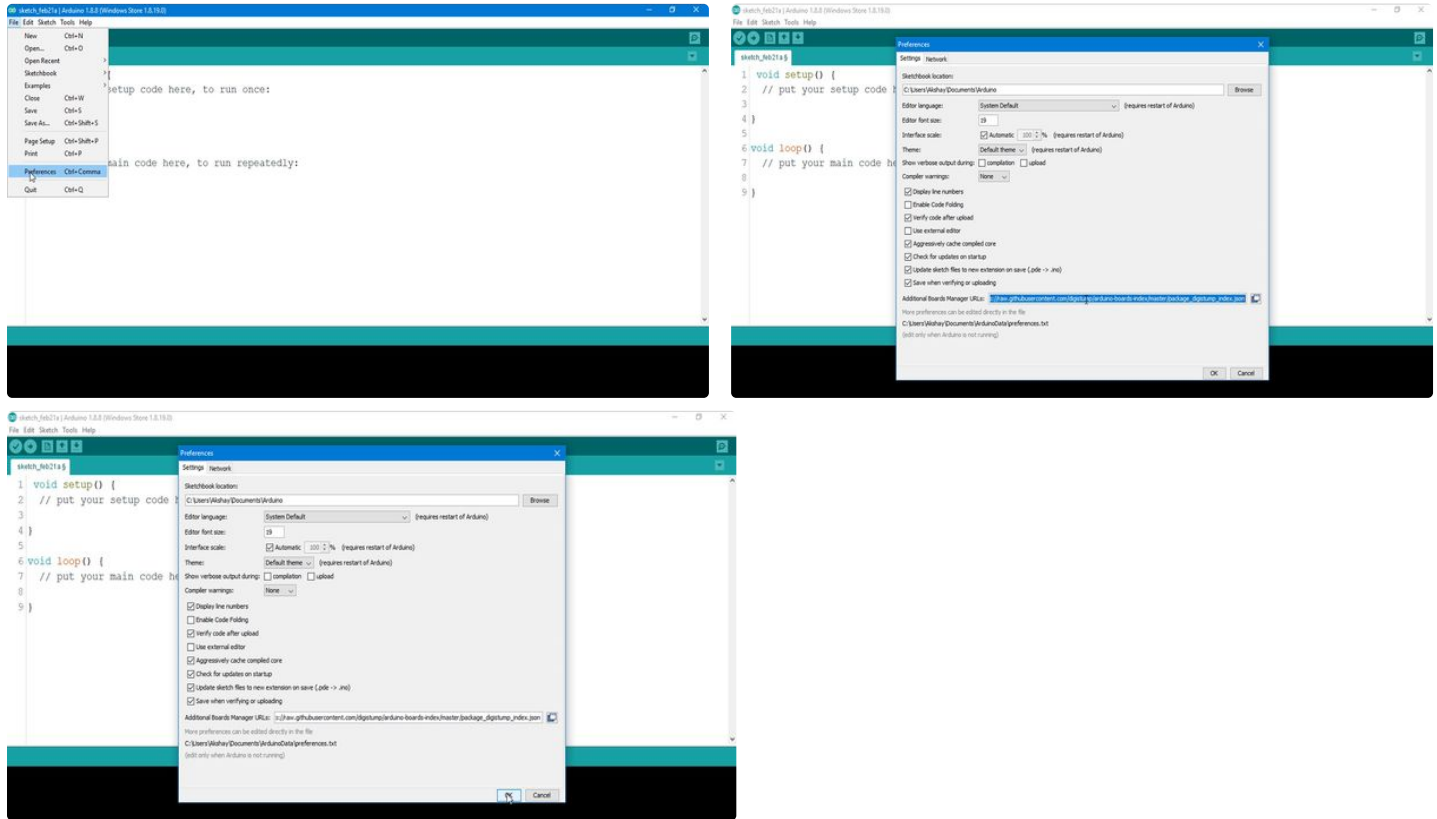
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Step 4: Edit Board URL

we need to add boards (DigiStump's Attiny85) into our Arduino IDE, to do so,

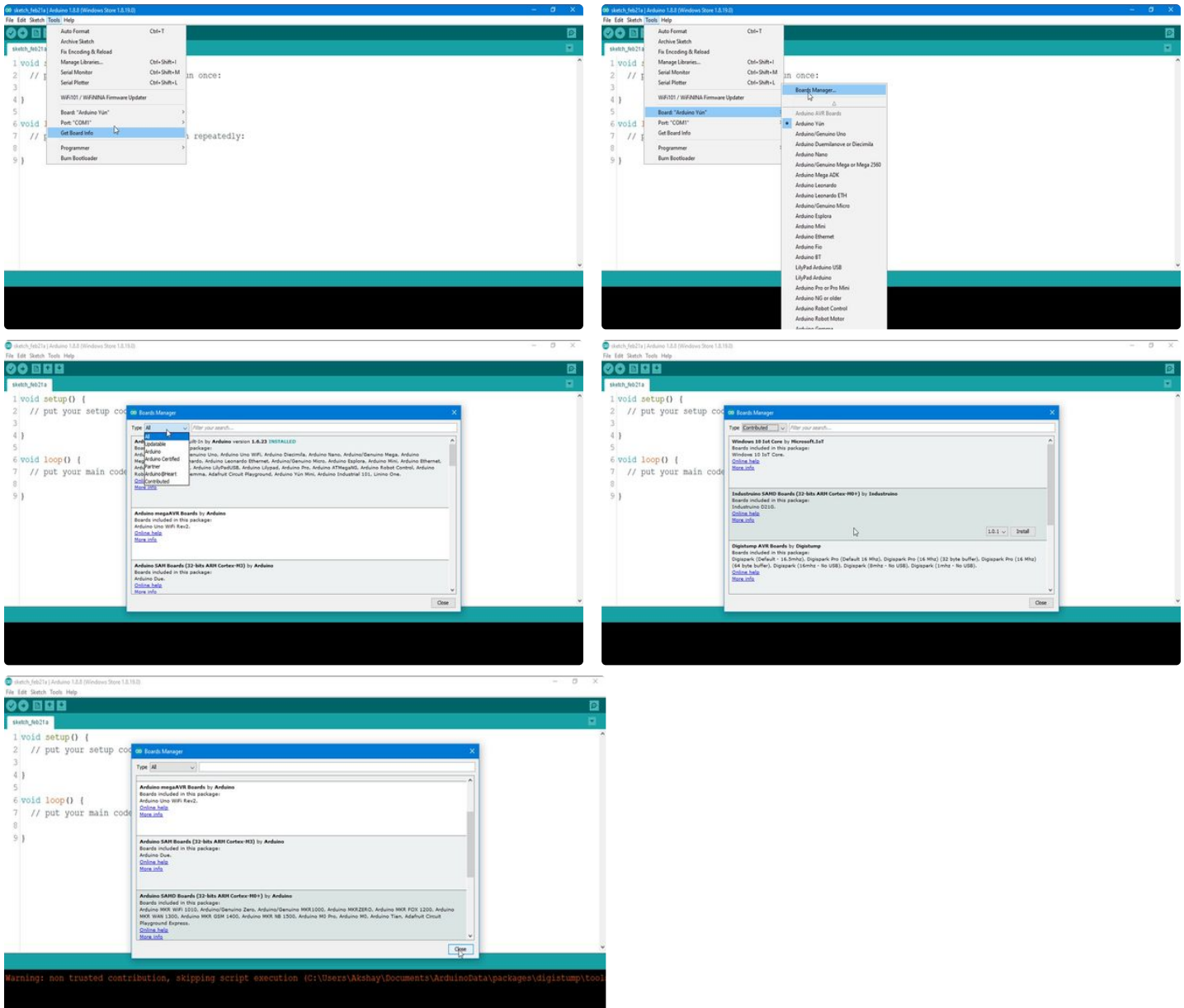
1. First **open** your Arduino **IDE**,
2. go to *Preference* and simply **paste** the Link "https://raw.githubusercontent.com/digistump/arduino-boards-index/master/package_digistump_index.json" on the **Additional Board Management URL**,
3. then simply Press **okay** and *Close* Preferences.



Step 5: Install DigiStump Board on IDE

to install Attiny Board Settings on Arduino IDE,

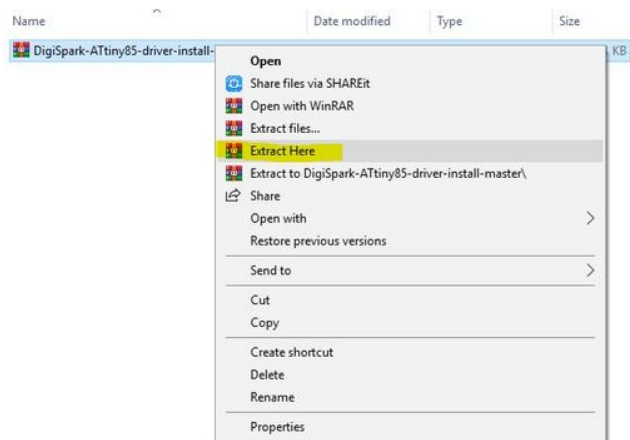
1. go to **board manager**, under Tools menu
2. select the **Contributed** type of boards Option
3. select "**DIGISTUMP AVR BOARDS**"
4. Click on **install** it!



Step 6: Install the Drivers

Now, install the drivers on your computer *Attached on this Step*

1. **Unzip** the Attached File.
2. click on **DPinst64.exe** to install the Drivers on your computer.
3. then **insert** the **Attiny85** board into your computer .



Name	Date modified	Type	Size
amd64	12/10/2016 2:07 AM	File folder	
x86	12/10/2016 2:07 AM	File folder	
cdc_digix	08/04/2016 1:21 PM	Security Catalog	8 KB
ChangeCDCSpeed	08/04/2016 1:21 PM	VBScript Script File	2 KB
digiserial	03/09/2014 10:45 ...	Security Catalog	9 KB
DigiSerial	03/09/2014 10:42 ...	Setup Information	3 KB
Digispark_Bootloader	08/04/2016 1:21 PM	Security Catalog	10 KB
Digispark_Bootloader	08/04/2016 1:21 PM	Setup Information	9 KB
digiusb	08/04/2016 1:21 PM	Security Catalog	11 KB
DigiUSB	08/04/2016 1:21 PM	Setup Information	8 KB
DigiX	08/04/2016 1:21 PM	Setup Information	4 KB
DPinst	08/04/2016 1:21 PM	Application	901 KB
DPinst64	08/04/2016 1:21 PM	Application	1,023 KB
Install Drivers	08/04/2016 1:21 PM	Application	1,487 KB
launcher	08/04/2016 1:21 PM	Application	1,416 KB
micronucleus	08/04/2016 1:21 PM	Application	82 KB
post_install	08/04/2016 1:21 PM	Windows Batch File	1 KB



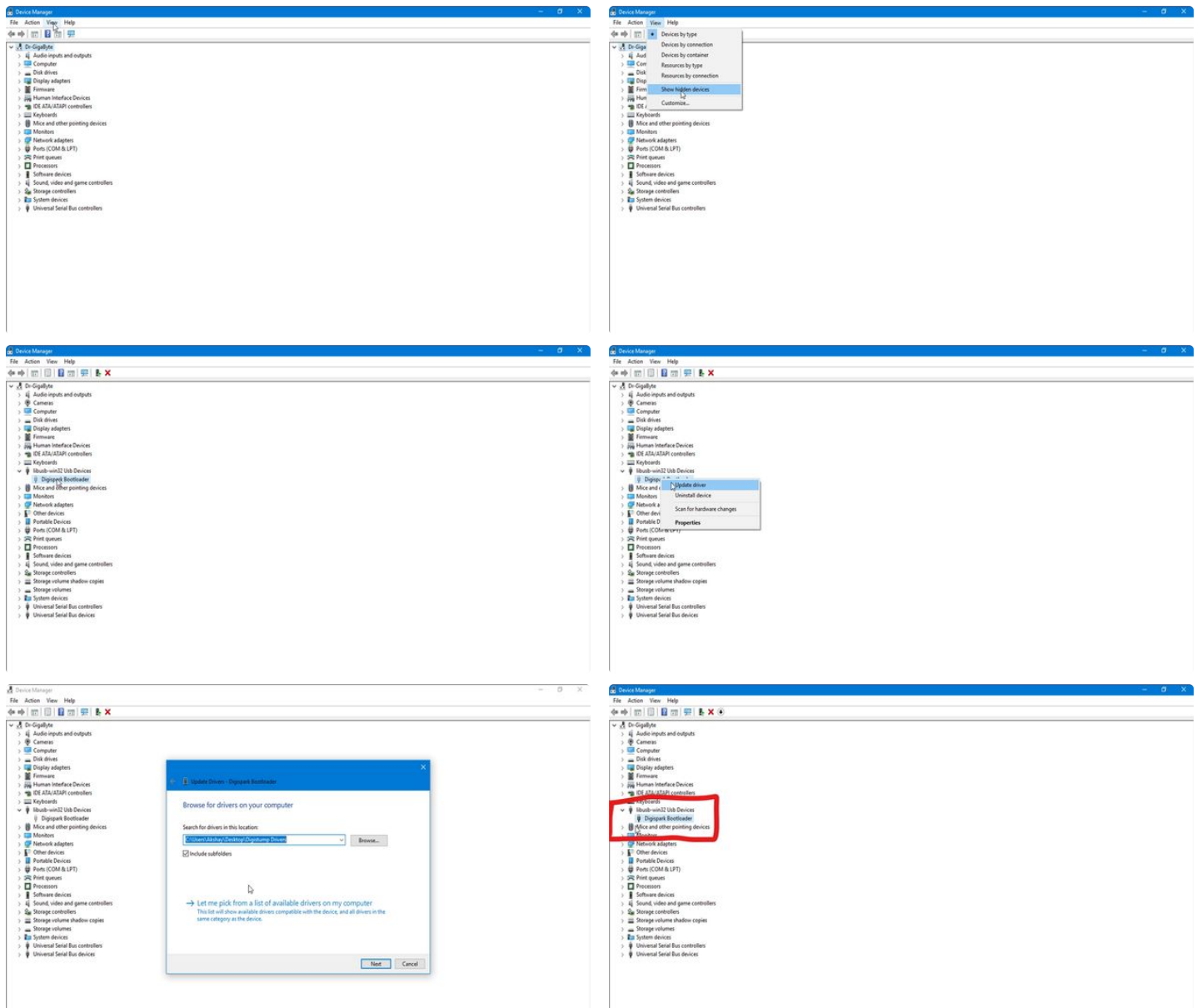
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Step 7: Troubleshooting (OPTIONAL)

sometimes your drivers are not installed correctly, in such (unusual) case, we will follow these steps

1. Open **Device Manager**,
2. Click on View Menu and select "**Show Hidden option**"
3. click on Device called "**LIBUSB-win**"
4. *Right Click* on **DigiSpark USB device** device and Click on **Update Device**
5. **browse** the **drivers** from our computer.
6. **select** the **location** of drivers stored on computer



Step 8: Write Test Code

- Now, let's run a test code to check our 'Attiny85' board,

we will simply try Blink code, which you can find attached in this step.

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<p>/* Blink Code by Akshay Momaya<br> * for Mission Critical Channel
 * ATTINY85 Tutorial for <Mission Critical><mission critical="">
 * subscribe for more arduino Tuorials and Projects
<a href="https://www.youtube.com/channel/UCM6rbuieQBBLFsxsxszWA85AQ?sub_confirmation=1">
https://www.youtube.com/channel/UCM6rbuieQBBLFsxs...</a>
*/</mission></p><p>void setup()
{
  pinMode(1, OUTPUT);
}</p><p>void loop()
{
  digitalWrite(1, HIGH);
  delay(1000);
  digitalWrite(1, LOW);
  delay(1000);
}

</p>
```



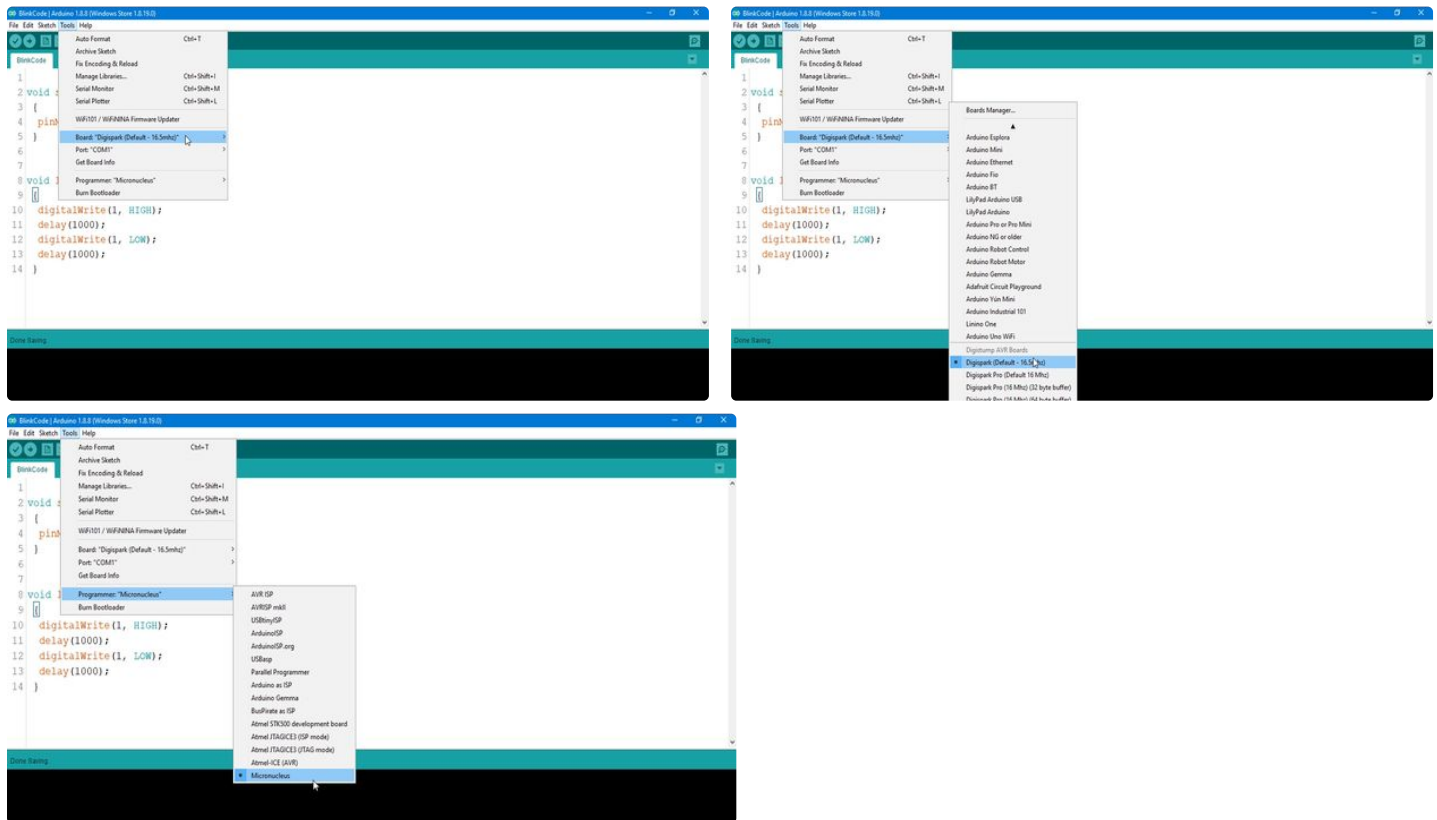
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Step 9: Select Proper Board Type

here, Selecting the Proper Code is the deal breaker of this entire Setup. to do so, we will

1. select “**Digispark (Default-16.5Mhz)**” from **Board Menu** in Tools.
2. select **programmer** as “**Micronucleus**” from Board Menu in Tools.



Step 10: Upload the Code

normally we would compile and Upload the code with our Arduino Connected, but here in Attiny85, we will

1. **Compile** the code
2. Press **UPLOAD** button
3. wait for Plug in device now... (will timeout in 60 seconds) this message.
4. **Connect** the Board Within **60 Seconds** .
5. wait for "**Micronucleus done. Thank you!**" Message From *compiler*.
6. *Enjoy!*

Output of this code will show blinking LED on Pin 1 on the Board

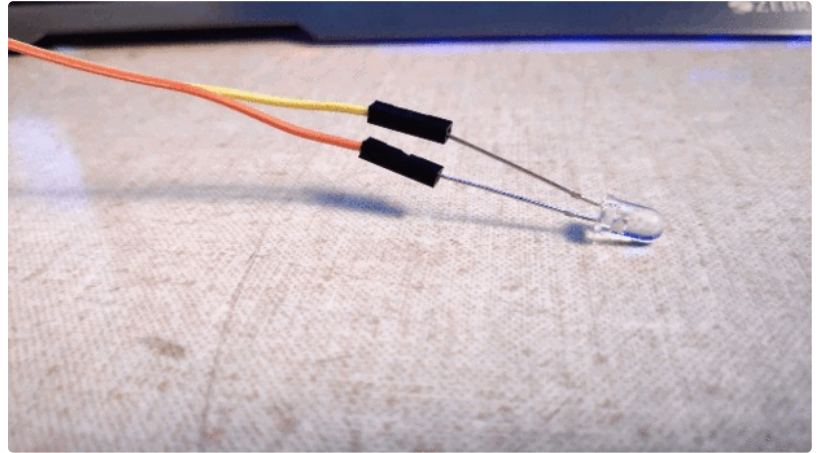
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Step 11: Conclusion / Future Scope

We will do much more projects and Experiments with this board in future,

like Shrinking your Circuit Size, making StandAlone Attiny85 Projects etc.

Don't forget to subscribe to Mission Critical for such Awesome Tutorials and Projects!



Or you can just buy the ATTiny85 chip (30 for \$1) and solder straight on but at \$2 for these boards on ebay why not.